

Anti-allergic activity of a Kampo (Japanese herbal) medicine “Sho-seiryu-to (Xiao-Qing-Long-Tang)” on airway inflammation in a mouse model

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Abstract

Effects of a Kampo (Japanese herbal) medicine “Sho-seiryu-to (SST, Xiao-Qing-Long-Tang in Chinese)”, which has been used for the treatment of allergic bronchial asthma clinically, were examined on ovalbumin (OVA)-sensitized allergic airway inflammation model (i.e., bronchial asthma) in a mouse. When SST was orally administered at 0.5 g/kg/day from day 1 to 6 days after OVA inhalation, SST reduced the OVA-specific IgE antibody titer in bronchoalveolar lavage (BAL) fluids at 7 days after the OVA inhalation. CD4⁺ T cells obtained from the mouse lung produced more interleukin (IL)-4 and IL-5 but less interferon (IFN)- γ than T cells from nonsensitized control animals. However, oral administration of SST reduced the production of IL-4 and IL-5 and the production of IFN- γ returned to the control level. In addition, the IL-4 level was increased in the BAL fluid of the OVA-sensitized animals compared to the nonsensitized control, while the IFN- γ levels decreased. SST reduced the IL-4 levels in the BAL fluids and returned the IFN- γ level to control levels. Nerve growth factor (NGF) was increased in the BAL fluids of the OVA-sensitized mice over that of nonsensitized mice, but oral administration of SST augmented the NGF levels to ~ 2 times higher than in the sensitized mice. Although lung cells obtained from sensitized mice produced higher levels of NGF than nonsensitized mice, oral administration of SST augmented the production of NGF by the lung cells even higher (~ 2 times more than cells from sensitized mice). Administration of anti-NGF antibody to the airway blocked the effects of SST. These results suggest that SST modulates Th1/Th2 balance in the lungs and augmentation of NGF in the lungs may be related to the effects of SST. Pinelllic acid (9S, 12S, 13S-trihydroxy-10E-octadecenoic acid), one component of the herbs of SST [Int. Immunopharmacol. 2 (2002) 1183], was purified from the tuber of *Pinellia ternata* Breitenbach. Oral administration of pinelllic acid (50 μ g/kg/day) also reduced the OVA-specific IgE antibody titer in BAL fluids from the sensitized mouse. This result suggests that pinelllic acid is one of active ingredient(s) in SST.

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1. Introduction

Allergic bronchial asthma is characterized by elevated serum IgE antibody level and inflammation of the airways with high levels of interleukin (IL)-4 and IL-5 in bronchoalveolar lavage (BAL) fluids and airways mucosa [1–3]. Constriction of airway smooth muscle and development of airway hyper-responsiveness (AHR) are also hallmarks of bronchial asthma [4,5]. Sho-seiryu-to (SST, Xiao-Qing-Long-Tang in Chinese) is a kind of Kampo (Japanese herbal) medicines and has been used clinically for the treatment of allergic rhinitis, bronchitis, bronchial asthma [6] and cold symptoms. In vitro antiallergic activities of SST include inhibition of histamine release and degranulation of mast cells [7,8], proliferation of eosinophils [9], growth and differentiation of basophils [10], cholinergic effects for nasal gland acinar cells [11], synthesis of tumor necrosis factor α by peripheral blood mononuclear cells [12] and contraction of bronchial smooth muscle [13,14]. In vivo antiallergic activities of SST include inhibition of passive cutaneous anaphylaxis (PCA) in rat [7,15] and it showed efficacies in animal models of allergic rhinitis [16–18] and allergic asthma [19,20]. However, few in vivo studies are available on the mode of actions of antiallergic activity of SST against bronchial asthma.

Nerve growth factor (NGF) is known as an endogenous factor that is essential for the survival, development, differentiation and maintenance of cholinergic neurons in the central nervous system. Recently, it has been reported that high NGF levels were found in the peripheral blood [21], BAL fluid [22] and bronchi [23] of patients with allergic diseases such as bronchial asthma. NGF levels were also elevated in mast cells [24,25], eosinophils [26], fibroblasts [27–29], T cells [30], B cells [31] and epithelial cells [32,33] from experimental allergic animal models. It has been reported that NGF enhanced the production of the tachykinin substance P [34]. These results suggest that NGF is involved in allergic diseases [35,36] such as bronchial asthma. However, effects of SST on NGF levels in allergic diseases have never been determined.

The active ingredient(s) responsible for the SST effects in the allergic airway inflammation model in mice has never been identified. Pinelllic acid (9*S*, 12*S*, 13*S*-trihydroxy-10*E*-octadecenoic acid) has been iden-

tified as an orally active adjuvant substance from the tuber of *Pinellia ternata* Breitenbach and one of the herbal components of SST [37]. Oral administration of pinelllic acid to mice given intranasal inoculation of influenza HA vaccine enhanced antiviral IgA antibody titers in nasal washes and BAL fluids, and antiviral IgG antibody titers in BAL fluids [37]. These results suggest that pinelllic acid modulate the production of antigen-specific antibodies in the respiratory tracts.

In present study, we elucidate the action of SST on the treatment of bronchial asthma. The in vivo effects of SST in mice were investigated by examining the production of allergen-specific IgE antibody, Th1/Th2 cytokines and NGF using an ovalbumin (OVA)-sensitized allergic airway inflammation model. In present study, the in vivo effect of pinelllic acid on production of allergen-specific IgE antibody was also investigated using the mouse model to help identify the active component in SST.

2. Materials and methods

2.1. Materials

Eight medicinal plants were used for preparation of a Kampo medicine, Sho-seiryu-to (SST). Pinelliae Tuber (tuber of *P. ternata*), Ephedrae Herba (stem of *Ephedra sinica*), Schizandrae Fructus (fruit of *Schizandra chinensis*), Paeoniae Radix (root of *Paeonia lactiflora*), Glycyrrhizae Radix (root of *Glycyrrhiza uralensis*) and Zingiberis Siccatur Rhizoma (scalded rhizoma of *Zingiber officinale*) were purchased from Uchida Wakan-yaku (Tokyo, Japan). Asiasari Radix (root of *Asiasarum sieboldii*) was obtained from Tsumura (Tokyo, Japan) and Cinnamomi Cortex (bark of *Cinnamomum cassia*) was from Tochimoto Tenkaido (Osaka, Japan). SST was prepared as previously described [38]. Briefly, a mixture of Pinelliae Tuber (6 g), Ephedrae Herba (3 g), Schizandrae Fructus (3 g), Cinnamomi Cortex (3 g), Paeoniae Radix (3 g), Asiasari Radix (3 g), Glycyrrhizae Radix (2 g) and Zingiberis Siccatur Rhizoma (1 g) was extracted with 600 ml of water. After the extract was centrifuged at 7500 rpm for 30 min, the supernatant was collected and lyophilized (yield, ~ 20%). Pinelllic acid was purified from Pinelliae Tuber as previously described [37]. Albiflorin, cinnamic acid,

cinnamic aldehyde, 6-gingerol, glycyrrhizin, liquiritin, paeoniflorin and schizandrin used for standards during HPLC separation were purchased from Wako Pure Chemical Industries (Osaka, Japan). Liquiritigenin and catechin were obtained from Extrasynthese (Genay, France).

2.2. Three-dimensional HPLC analysis of SST

SST extract (10 mg) was dissolved in water (1 ml) and mixed with vortex mixer and sonicated. The suspension was filtered using a DISMIC-13HP PTFE membrane filter (0.45 μ m, Advantec Toyo). Analysis of the filtrate was performed by HPLC. Conditions of HPLC analysis were as follows. HPLC, Hewlett Packard, HP Kayak XM600 Agilent 1100 Series; detector, photodiode-array detector; column, TSK gel ODS-80Ts (4.6 $\times$ 250 mm; Tosoh, Tokyo, Japan); solvent phase, a linear gradient of 10

mM phosphoric acid–acetonitrile (95:5 $\rightarrow$ 5:95) for 60 min at a flow rate of 0.8 ml/min and column temperature of 40 °C. The three-dimensional HPLC profile of an aqueous SST extract is shown in Fig. 1. The analysis based on ultraviolet (UV)-absorption clearly showed the presence of the following major constituents in SST: liquiritin, liquiritigenin and glycyrrhizin (originating from *Glycyrrhizae Radix*); paeoniflorin and albiflorin (*Paeoniae Radix*); schizandrin (*Schizandrae Fructus*); cinnamic aldehyde and cinnamic acid (*Cinnamomi Cortex*); catechin (*Ephedrae Herba*); and 6-gingerol (*Zingiberis Siccatum Rhizoma*).

2.3. Mice

Five-weeks-old female BALB/c mice were purchased from Japan SLC (Hamamatsu, Japan) or CLEA Japan (Tokyo, Japan). The animals were

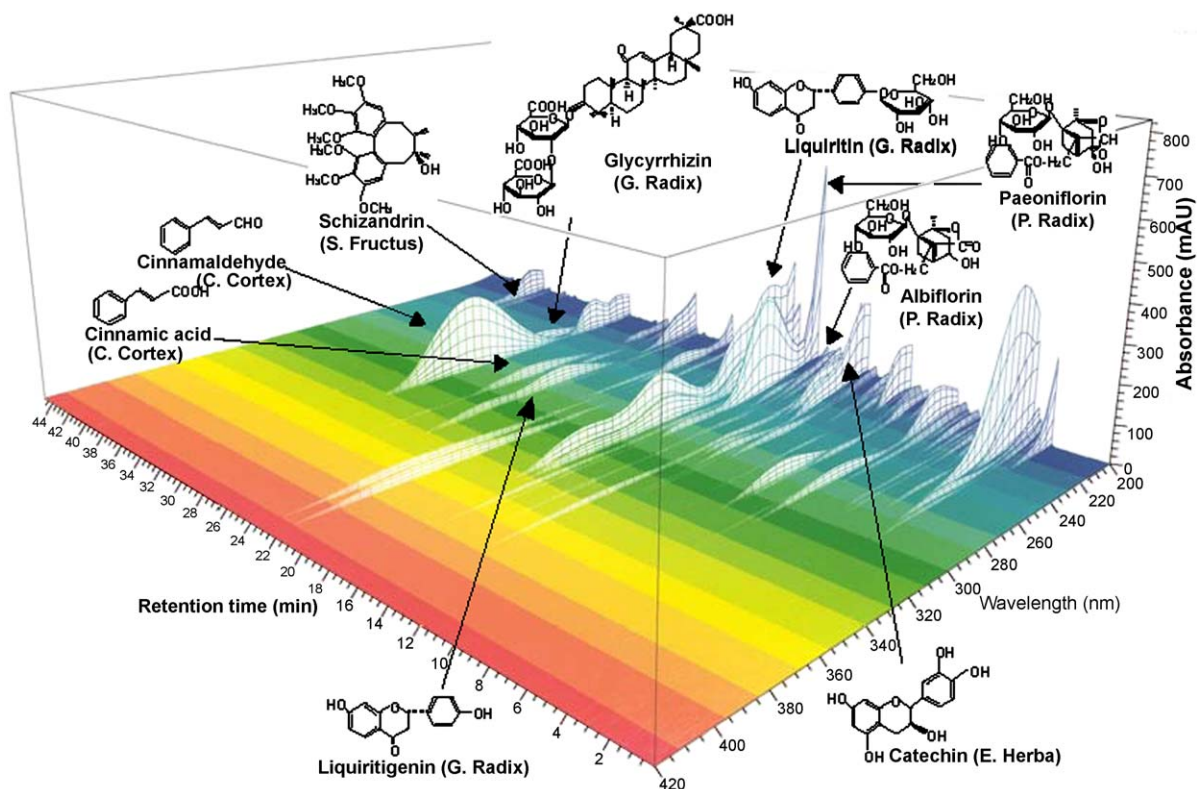


Fig. 1. Chemical profile of the hot water extract from Sho-seiryu-to (SST) as analyzed by 3-dimensional HPLC.

housed in plastic cages in an air-conditioned room at 23 ± 2 °C with a relative humidity of $55 \pm 10\%$ under a 12-h light–dark cycle, fed a standard laboratory diet and given water ad libitum. Animal experiments were performed in accordance with the animal care guideline of Kitasato Institute and Kitasato University.

2.4. Preparation of allergic airway inflammation model mice

The mouse allergic airway inflammation model was prepared according to methods previously described [38–40]. Mice, 6 weeks old, were sensitized by an intraperitoneal injection of 0.5 ml of alum-precipitated antigen containing 8 µg of OVA (grade VI, Sigma) absorbed to 2 mg of alum (aluminum hydroxide hydrate gel, LSL) in saline vehicle. A booster injection of this alum-OVA mixture was given 5 days later. Nonsensitized control animals received alum only. Seven days after sensitization, the mice were exposed to provocation of antigen. For the challenge of antigen, the mice were placed in a chamber (Sanki Kagaku Kogei) and were exposed twice to aerosolized OVA (0.5%) in PBS for 1 h at 4 h intervals. The aerosolized OVA was produced by an ultrasonic nebulizer (Model NE-U12, OMRON).

2.5. Preparation of lung cells

Mice were anesthetized with diethyl ether and then bled from the heart with a syringe. After exsanguination, the lung was removed and incubated for 60 min at 37 °C with 4 mg/ml of collagenase H (Roche Diagnostics) in 2 ml of RPMI 1640 medium (Sigma) supplemented with penicillin (100 U/ml) and streptomycin (100 µg/ml) on a rotator. To dissociate tissue into single cells, collagenase-treated lung cells were dispersed by tapping gently with rubber rod on a 150-gauge sterile stainless sieve in the plastic dish (6-cm diameter). The cell suspensions were passed through 200-gauge sterile stainless sieve, washed three times with the medium and resuspended in 6 ml of the culture medium. The cells were layered on top of a Percoll (Amersham Biosciences) density gradient, prepared with 2 ml of 40% and 2 ml of 72% Percoll solutions in PBS. After centrifugation at 2000 rpm for 15 min at room temperature, highly viable (>95%)

lung lymphoid cells were recovered from the interface between the 40% and 72% Percoll solution. The lung cells [2×10^7 cells/ml in RPMI 1640 containing 5% fetal bovine serum (FBS)] were cultured in the 96-well culture plate (BD Falcon) at 37 °C under a 5% CO₂ atmosphere.

2.6. Production of cytokines by lung CD4⁺ T cells

CD4⁺ T cells were prepared from the single-cell suspension of mouse lung cells and collected using Dynabeads® mouse CD4 and DETACHaBEAD mouse CD4, by a magnetic cell sorting system (DynaL Biotech, Oslo, Norway) according to the manufacturer's instructions. Murine lung CD4⁺ T cell suspension (2.5×10^6 cells/ml in RPMI 1640 containing 10% FBS) was incubated in the 96-well culture plate at 37 °C for 7 days under a 5% CO₂ atmosphere. The concentrations of interleukin (IL)-4, IL-5 and interferon (IFN)-γ in the supernatant were measured by ELISA systems (Endogen, Woburn, MA, USA; Sensitivity: <5, <5 and <10 pg/ml for IL-4, IL-5 and IFN-γ, respectively).

2.7. Specimens

Mice were anesthetized with diethyl ether and then bled from the heart with a syringe. Serum was separated from the blood by centrifugation and used for titration of antibody. After bleeding, mice were incised ventrally along the median line from the xiphoid process to the point of the chin. The trachea with lung attached was taken out and washed twice by inflating the lung through the trachea with 2 ml of PBS containing 0.1% bovine serum albumin (BSA). The BAL fluid was used after the removal of cellular debris by centrifugation at 2500 rpm for 20 min at 4 °C.

2.8. ELISA for anti-OVA IgE and IgG1 antibodies

The OVA-specific IgE and IgG1 antibody titers in BAL fluid and serum were measured by the modified fluorometric reverse (antibody-capture) ELISA [37,41]. Briefly, the wells of a 96-well ELISA plate (Immulon® 4 HBX, Thermo Labsystems; Franklin, MA, USA) were coated with 100 µl of anti-mouse IgE or IgG1 mAb (BD Pharmingen, San Diego, CA, USA;

1 µg/ml) in 50 mM carbonate–bicarbonate buffer (pH 9.5) containing BSA (10 µg/ml). The plate was incubated at 37 °C for 3 h. After the solution was removed, the blocking solution, 1% skim milk in PBS, was placed in each well (300 µl) and incubated at 37 °C for 1 h. The plates were washed four times with PBS containing 0.05% Tween 20 (PBS–Tween). Serial dilutions of BAL fluid or serum with Super-Block® blocking buffer in PBS (Pierce, Rockford, IL, USA) containing 0.05% Tween 20 (100 µl) were added to the wells. After being sealed with adhesive tape, the plates were incubated overnight at room temperature. After washing with PBS–Tween, 1 µg/ml of biotinylated OVA in the blocking solution was added to each well (100 µl). The plates were incubated at room temperature for 1 h with shaking on a microplate mixer. After washing the wells, streptavidin-β-galactosidase conjugate (Invitrogen, Carlsbad, CA, USA), diluted to 1:1000 with the blocking solution, was added (100 µl) and incubated at room temperature for 1 h with shaking. After the final wash, 0.1 mM 4-methylumbelliferyl-β-D-galactoside (Sigma) in buffer A (10 mM sodium phosphate buffer, pH 7.0, containing 0.1 M NaCl, 1 mM MgCl₂, 0.1% BSA and 0.1% NaN₃) was added to each well (100 µl). The plates were sealed with tape and then incubated at 37 °C for 2 h. The enzyme reaction was stopped by the addition of 100 µl of 0.1 M glycine–NaOH (pH 10.3), and the fluorescence of the 4-methylumbelliferone measured (ex. 355 nm, em. 460 nm) using a Fluoroskan II spectrofluorometer (Thermo Labsystems). Endpoint titers of OVA-specific IgE and IgG1 antibodies were expressed as reciprocal log₂ titers.

2.9. ELISA for NGF

The concentrations of NGF in BAL fluid, and cell culture supernatant were measured using an ELISA kit (NGF Emax® ImmunoAssay System) according to the manufacturer's instruction (Promega, Madison, WI, USA).

2.10. Intranasal treatment by anti-NGF antibody

Intranasal treatments of mice by anti-NGF antibody were performed according to the method of Braun et al. [42] with minor modification. Briefly, mice were anesthetized by an intraperitoneal injection

of sodium amobarbital (0.25 ml of a saline solution of 11 mg/ml; Sigma). The mice were administered intranasally, 25 µl (50 µg/kg) of rabbit anti-mouse NGF (IgG fraction) polyclonal antibody (Chemicon International, Temecula, CA, USA) or isotype control rabbit IgG (Southern Biotechnology Associates, Birmingham, AL, USA) at 3 h before and 1, 3 and 6 days after the challenge of allergen through airway.

2.11. Statistical analysis

All results were presented as mean ± S.E.M. from the number of experiments indicated. Differences between two groups were analyzed by Student's *t*-test. Differences among three groups or more were analyzed by Dunnett's test of either parametric or nonparametric type. *P* < 0.05 was considered to be significant.

3. Results

3.1. Effect of SST on production of anti-OVA antibodies

Effect of oral administration of SST on the anti-OVA IgE and IgG1 antibody titers was examined in the serum and BAL fluid of OVA-sensitized mice as an airway inflammation model. SST was administered orally at 0.5 g/kg/day from day 1 to 6 days after the OVA inhalation, and serum and BAL fluid were prepared at 7 days after the inhalation. Anti-OVA IgE antibody titers in the serum (Fig. 2A) and BAL fluid (Fig. 2B) of the sensitized mice were significantly greater than the titers in the nonsensitized control. Anti-OVA IgG1 antibody titers in the serum and BAL fluid of OVA-sensitized mice were also significantly greater than titers from control mice (data not shown). When SST was administered to the OVA-sensitized mice, anti-OVA IgG1 antibody titers in the serum and BAL fluid were not affected (data not shown). Anti-OVA IgE antibody titers in serum were also unchanged by the administration of SST (Fig. 2A), but the antibody titer in the BAL fluid decreased significantly in comparison with that of water-administered sensitized mice (Fig. 2B). These results suggest that SST specifically decreases

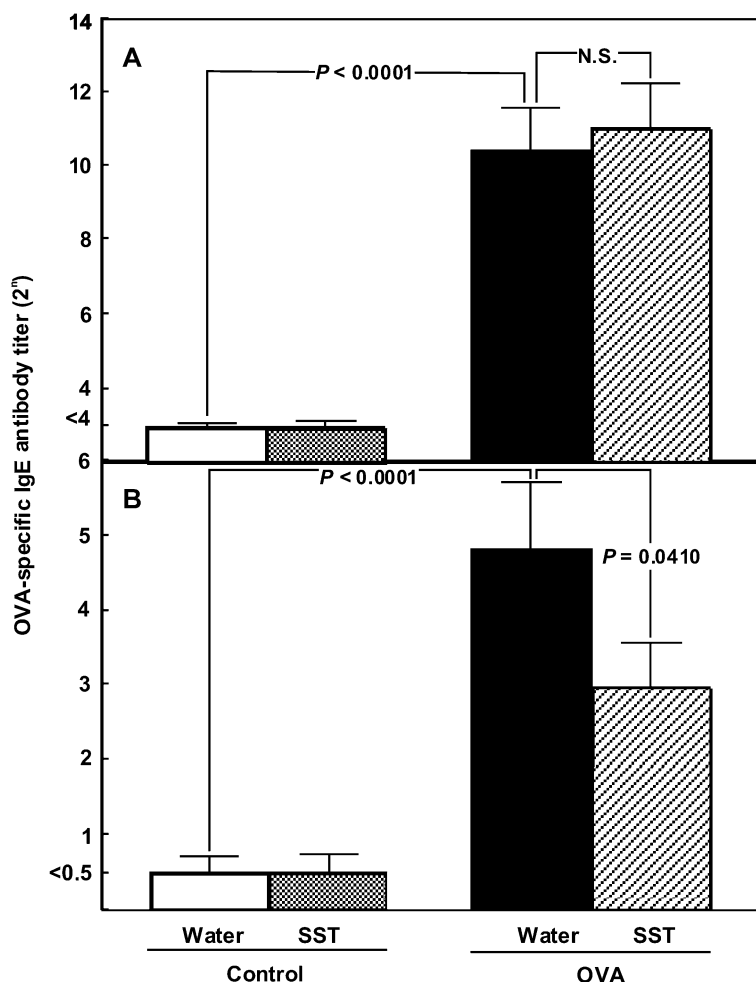


Fig. 2. Effect of SST on OVA-specific IgE antibody titers in OVA-sensitized mice was measured. OVA-sensitized BALB/c mice were treated p.o. with SST (0.5 g/kg) or water five times from day 1 to 6 days after OVA exposure. OVA-specific IgE antibody titers in serum (A) and BAL fluid (B) were determined at 7 days after OVA exposure as described in Materials and Methods. Values represent the mean $\pm$ S.E. ($n=7$). NS=not significant.

OVA-specific IgE antibody levels in BAL fluid but not in serum of the sensitized mice.

3.2. Effects of SST on production of cytokines in lung CD4⁺ T cells and cytokine levels in BAL fluids

Effects of oral administration of SST on the production of cytokines from CD4⁺ T cells were studied in the lung of OVA-sensitized mice. Production of IL-4 was not detectable in CD4⁺ T cells prepared from the lung of nonsensitized control mice, but it was

markedly increased in OVA-sensitized mice (Fig. 3A). CD4⁺ T cells from the lung of OVA-sensitized mice produced IL-5 at ~ 12 times higher levels than cells from nonsensitized control animals (Fig. 3B). When SST was administered orally to the OVA-sensitized mice, the productions of IL-4 and IL-5 from lung CD4⁺ T cells were reduced to the levels of nonsensitized control (Fig. 3A and B). Production of IFN- γ by CD4⁺ T cells from the lung of OVA-sensitized mice was significantly lower than that by cells from nonsensitized mice (Fig. 3C). When SST

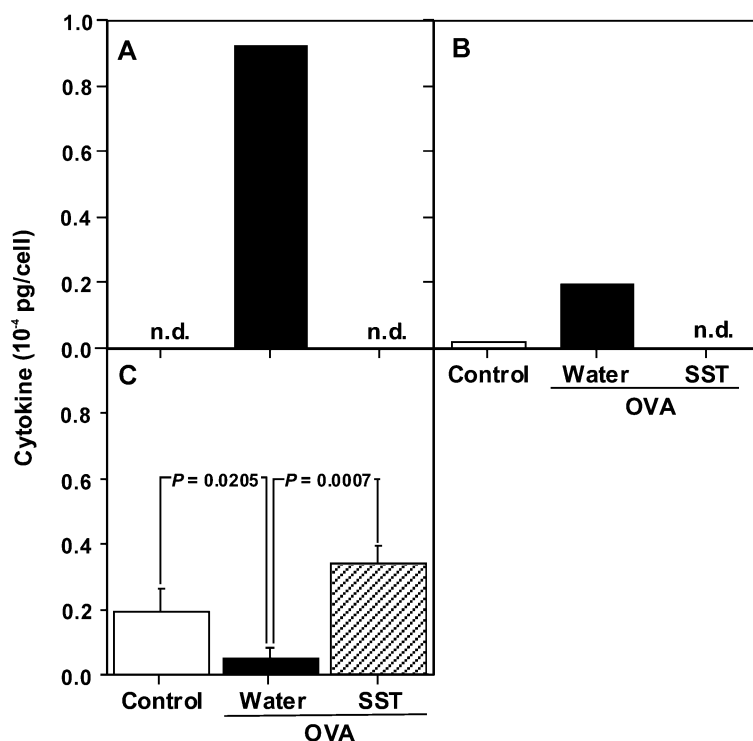


Fig. 3. Effect of SST on the production of cytokines by lung CD4⁺ T cells from OVA-sensitized mice was measured. OVA-sensitized mice were treated with SST or water as described in legend of Fig. 2. CD4⁺ T cells were prepared from lung at 7 days after OVA exposure and were pooled each group ($n=10$), and contents of IL-4 (A), IL-5 (B) and IFN- γ (C) in culture supernatant of the cells were determined as described in Materials and Methods. Values in (C) represent the mean $\pm$ S.E. of three separate experiments. nd = not detectable.

was administered to the OVA-sensitized mice, the production of IFN- γ from lung CD4⁺ T cells was significantly higher than that of cells from the OVA-sensitized mice (Fig. 3C). These results suggest that lung CD4⁺ T cells from OVA-sensitized mice are type 2 helper T (Th2) cell dominant, and SST modulates the CD4⁺ T cells to Th1 cell dominant.

Effects of SST on the cytokine levels in BAL fluids of OVA-sensitized mice were studied. IL-4 levels increased significantly in BAL fluids from OVA-sensitized mice compared to IL-4 levels in BAL fluids from nonsensitized control mice (Fig. 4A). When SST was administered orally to the OVA-sensitized mice, the IL-4 level was reduced significantly (Fig. 4A). IFN- γ levels were reduced in the BAL fluids from OVA-sensitized mice compared to levels in BAL fluids from non-sensitized mice (Fig. 4B). When SST was administered to the OVA-sensitized mice, the IFN- γ level increased significantly (Fig. 4B).

These results also suggest that cells in the bronchoalveolar (BA) cavity of OVA-sensitized mice are Th2 cytokine dominant and that SST modulates the BA cavity cells which are Th1 cytokine dominant.

3.3. Effect of SST on NGF level in OVA-sensitized mice

It has been reported that the NGF levels in BAL fluid and blood of asthmatic patients is enhanced after segmental allergen challenge [21,22]. However, the effect of SST on NGF levels in asthmatic patients has not been evaluated. Therefore, the effect of oral administration of SST on NGF levels in BAL fluids of OVA-sensitized mice was examined. Concentration of total NGF in BAL fluids from OVA-sensitized mice was significantly higher than that of nonsensitized controls (Fig. 5). When SST was administered to the OVA-sensitized mice, the NGF level in BAL fluids

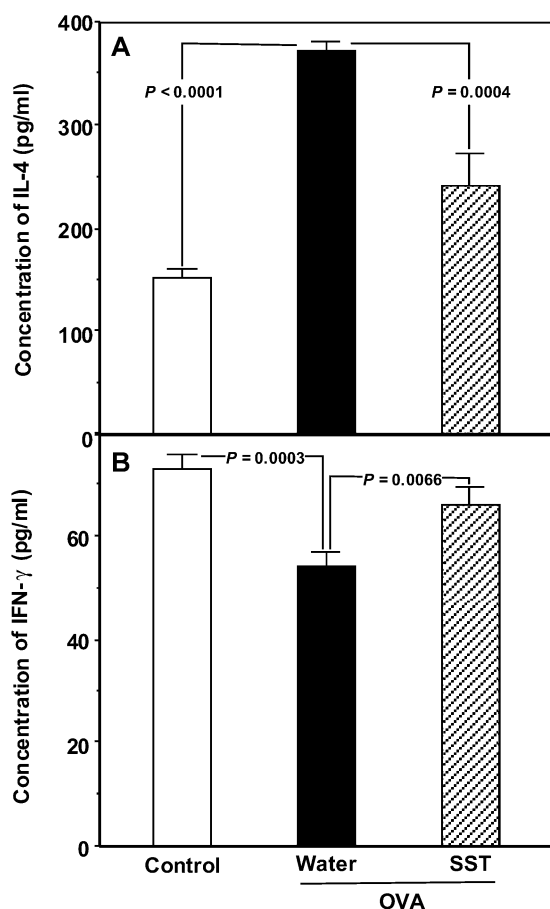


Fig. 4. Effect of SST on cytokine levels in the BAL fluid of OVA-sensitized mice was measured. OVA-sensitized mice were treated with SST or water as described in legend of Fig. 2. Concentrations of IL-4 (A) and IFN- γ (B) in BAL fluids were determined as described in Materials and Methods. Values represent the mean $\pm$ S.E. ($n = 7$).

was augmented significantly in comparison with that of OVA-sensitized mice followed by administration of water (Fig. 5). These results suggest that the oral administration of SST elevates NGF levels in the BA cavity of OVA-sensitized mice.

3.4. Effect of SST on secretion of NGF from lung cells

To assess which lung cell types were a source of NGF production, lymphoid cells were prepared from the lungs of non-sensitized and OVA-sensitized mice, respectively. The concentration of NGF increased

time dependently in culture supernatant of lung cells prepared from OVA-sensitized mice, and NGF levels were significantly higher than those from non-sensitized control animals at 3 and 12 h, respectively (Fig. 6). When lung cells obtained from the SST treated OVA-sensitized mice were cultured, levels of NGF in the supernatant were significantly higher than those from OVA-sensitized water-administered mice at 3, 9 and 12 h, respectively (Fig. 6). These results indicated that SST up-regulated the production of NGF in the lung cells of OVA-sensitized mice.

3.5. Effect of anti-NGF antibody on activities of SST

To assess the contribution of NGF on SST effects such as reduction of anti-OVA IgE antibody and IL-4 levels, and recovery of IFN- γ levels in the BAL fluid, anti-NGF antibody was administered intranasally to OVA-sensitized mice. BAL fluids were prepared from the mice at 7 days after the final challenge of OVA to the airway. When anti-NGF antibody was administered to the SST-treated OVA-sensitized mice, the anti-OVA IgE antibody titer in the BAL fluid increased significantly in comparison to the titers of SST-treated sensitized mice administered nonimmune IgG (Fig. 7A). Oral administration of SST reduced the concentration of IL-4 in BAL fluid of OVA-sensitized mice significantly, while administration of

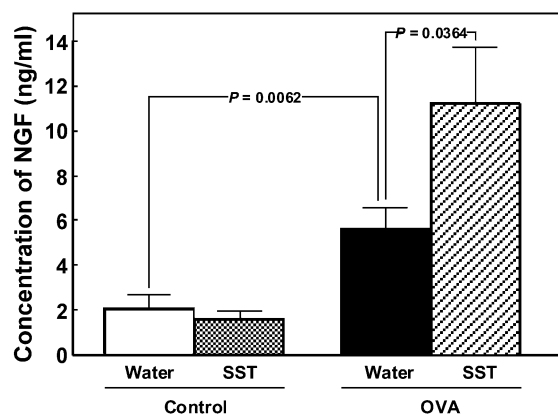


Fig. 5. Effect of SST on NGF level in the BAL fluid of OVA-sensitized mice was measured. OVA-sensitized mice were treated with SST or water as described in legend of Fig. 2. Concentration of NGF in BAL fluid was determined as described in Materials and Methods. Values represent the mean $\pm$ S.E. ($n = 10$).

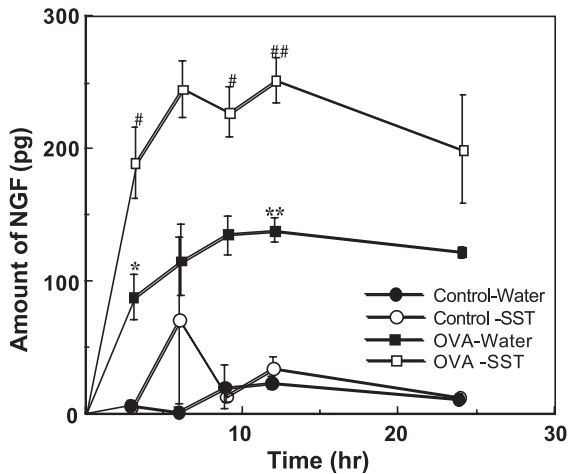


Fig. 6. Effect of SST on production of NGF from lung cells of OVA-sensitized mice was measured. OVA-sensitized mice were treated with SST or water as described Fig. 2. Lung cells were prepared at 7 days after OVA exposure as described in Materials and Methods and each group was pooled ($n = 10$). Kinetics of NGF contents in culture supernatant of the lung cells was determined for 24 h. Each plot represents the mean $\pm$ S.E. of three separate experiments. * $P < 0.05$, ** $P < 0.001$, compared with nonsensitized/water-administered mice. # $P < 0.01$, ## $P < 0.001$, compared with OVA-sensitized/water-administered mice.

anti-NGF antibody to the OVA-sensitized/SST-treated mice increased IL-4 levels significantly (Fig. 7B). The concentration of IFN- γ showed the tendency to increase in BAL fluid of SST-treated OVA-sensitized mice, but these levels decreased significantly after administration of anti-NGF antibody (Fig. 7C). These results suggest that administration of anti-NGF antibody to the respiratory tracts prevents the effects of SST in reducing OVA-specific IgE antibody and IL-4 levels and the induction of IFN- γ in BAL fluids.

3.6. Effect of pinelllic acid on production of anti-OVA IgE antibody

The effect of orally administered pinelllic acid on the anti-OVA IgE antibody titers in the serum and BAL fluid of OVA-sensitized mice was examined. Pinelllic acid was administered orally at 50 $\mu\text{g/kg/day}$ from day 1 to 6 days after the OVA inhalation, and serum and BAL fluid were prepared at 7 days after the inhalation. Anti-OVA IgE antibody titers in serum did

not decrease after the administration of pinelllic acid (Fig. 8A) but the antibody titer in the BAL fluid was significantly decreased in comparison with the antibody titer in BAL fluid from water-administered sensitized mice (Fig. 8B). These results suggest that pinelllic acid specifically decreases OVA-specific IgE antibody levels in BAL fluid but not in serum of the sensitized mice.

4. Discussion

SST has been widely used clinically for the treatment of allergic diseases including bronchial asthma and allergic rhinitis. The mechanism of SST action in allergic diseases has been studied in vitro and in vivo. However, its mode of action is still not fully elucidated for bronchial asthma. It has been reported that bronchial asthmatic patients [43] and animal models [44] are shown to be Th2-cell-dominant. In the present study, SST reduced the productions of Th2 cell-associated cytokines, IL-4 and IL-5, and restored production of the Th1 cell-associated cytokine, IFN- γ , from CD4 $^{+}$ T cells prepared from lungs of sensitized mice compared to that in nonsensitized control mice. SST also reduced the IL-4 levels and restored the IFN- γ levels in BAL fluid from the lungs of sensitized mice. The changes in these cytokine levels in BAL fluid were no more dramatic than their productions from CD4 $^{+}$ T cells. It may be that the cytokines detected in BAL fluid originated from not only lung CD4 $^{+}$ T cells but also from other cells or serum.

Anti-OVA IgE antibody levels were reduced in the BAL fluids of the sensitized mice after oral administration of SST. It has been reported that Th2 cells encourage production of IgE antibody [45–47]. These results indicate that SST has a function that modulates the proliferation of lung Th cells from Th2-cell-dominant to Th1-cell-dominant in our airway inflammation model of mice. However, SST did not reduce anti-OVA IgE antibody level in the serum of these sensitized mice. This result suggests that SST shows antiallergic effects on immune systems in bronchoalveolar (BA) cavities but not on systemic immune responses. Recently, Ikeda et al. [18] reported that OVA-specific IgE level in the serum and production of IL-4 in the splenocytes was suppressed by a pharmaceutical preparation of Sho-

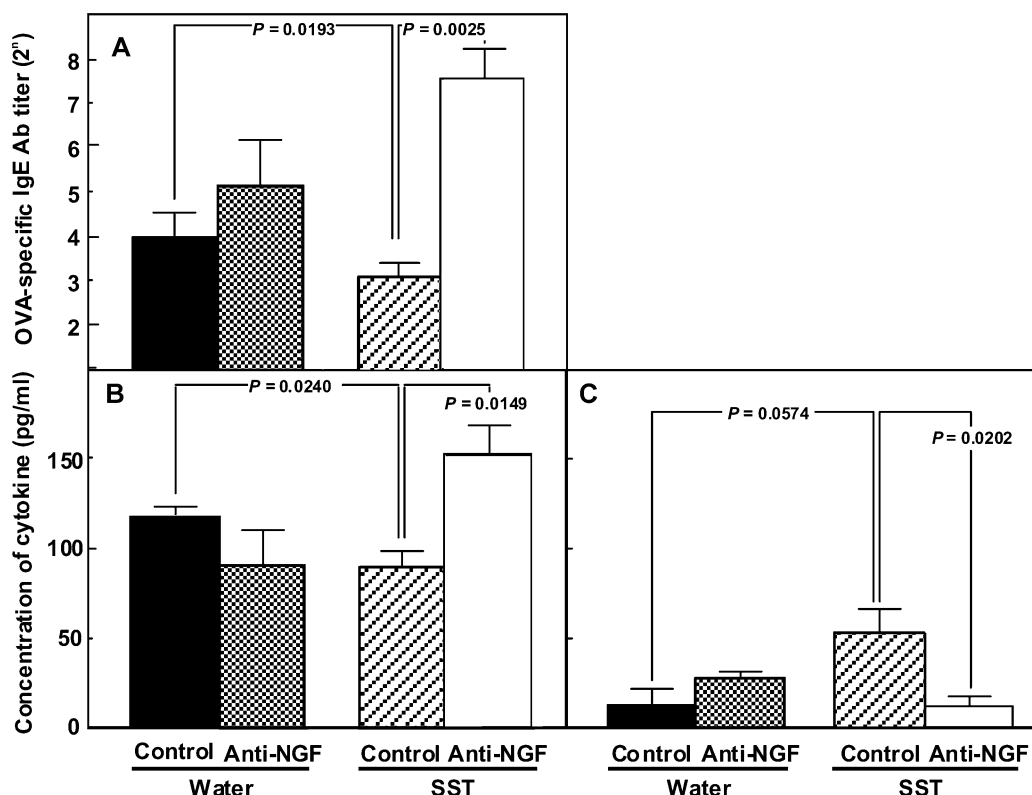


Fig. 7. Effects of intranasal anti-NGF antibody treatment on OVA-specific IgE antibody titer and concentration of cytokines in BAL fluid of OVA-sensitized and SST-administered mice were measured. OVA-sensitized and SST-administered mice were treated with anti-NGF antibody (50 μ g/kg) or nonimmune IgG (control) at 3 h before and 1, 3 and 6 days after the challenge of allergen to the airway. OVA-specific IgE antibody titer (A) and concentration of IL-4 (B) or IFN- γ (C) in BAL fluid were determined at 24 h after the last treatment by anti-NGF antibody as described in Materials and Methods. Values represent the mean $\pm$ S.E. ($n=7-8$).

seiryu-to (TJ-19) in the nasal allergic mouse model. In their study, TJ-19 showed significant effects when the dose of 1.0 g/kg/day was administrated for 28 days after first sensitization in the nasal allergic mouse model [18]. However, in the present study, significant effects of SST were observed in the BA cavities when it was administered at 0.5 g/kg/day for only 5 days after the challenge of allergen in our airway inflammation model. These results suggest that the difference between the previous report and the present study may be caused by the use of different allergic models or timing and dose of the administration of Sho-seiryu-to.

In the present study, SST enhanced NGF levels by about twofold more in the lung of the sensitized mice. Several hallmarks of bronchial asthma were reduced

by SST, such as anti-OVA IgE antibody in BAL fluids and Th2 cell dominance on CD4⁺ T cells of the lung. However, treatment of anti-NGF antibody prevented the antiallergic effects of SST in the BAL fluid. These findings suggest that NGF in the respiratory tracts is related to the pharmacological effects of SST on the treatment of bronchial asthma. It has been reported that NGF is overexpressed in the infiltrated cells on the lung and is markedly increased in the serum and BAL fluids of OVA-sensitized mice [42,48]. These observations suggest that NGF relates to the development of allergic responses in the airway and lung. It has also been reported that nasal application of anti-NGF to OVA-sensitized mice reduced IL-4 but did not affect IFN- γ in BAL fluid when determined within 24 h after the last challenge of

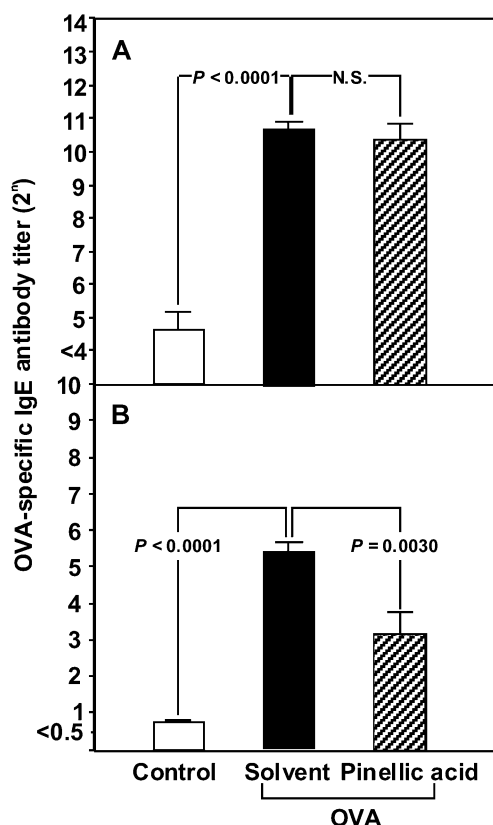


Fig. 8. Effect of pinellic acid on OVA-specific IgE antibody titers in OVA-sensitized mice was measured. OVA-sensitized mice were treated p.o. with pinellic acid (50 μ g/kg) or the solvent from day 1 to 6 days (i.e., five times) after OVA exposure. OVA-specific IgE antibody titers in serum (A) and BAL fluid (B) were determined at 7 days after OVA exposure as described in Materials and Methods. Values represent the mean $\pm$ S.E. ($n = 7$). NS = not significant.

aerosolized allergen [43,49]. In the present study, the parameters were measured at 7 days after the challenge of allergen, which may be at the repair stage of airway inflammation. It has been reported that NGF plays an important role in repair of tissue [29,49,50]. SST may modulate the effect of NGF to enhance the processes of repair that takes place after or concomitantly with the tissue damage in the respiratory tracts caused by inflammation.

In the present report, we suggested that antiallergic effects of SST might be related to the augmentation of NGF in the lung of sensitized mice (i.e., in allergic airway inflammation model). To further understand

the mechanisms of SST on pharmacological activities, one must clarify the mechanism for upregulation of NGF in the lung and elucidate the origin and types of the cells augmented by NGF.

In order to identify an active ingredient of the SST mixture, we examined the effects of pinellic acid. Pinellic acid has been purified from one of component herbs of SST, *Pinelliae Tuber* [37]. In the present paper, anti-OVA IgE antibody levels were reduced in the BAL fluids of the OVA-sensitized mice by oral administration of pinellic acid but not in the serum. Oral administration of SST also reduced anti-OVA IgE antibody levels in the BAL fluids of these mice but not in the serum. These results indicated that pinellic acid might be one of the active compounds in SST causing a reduction of anti-OVA IgE antibody in the BAL fluids of the sensitized mice. However, involvement of molecules other than pinellic acid from the various component herbs of SST must be anticipated. Certain actions of Kampo medicines are suggested to occur from a combinational effect of the multiple constituents in the medicine.

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